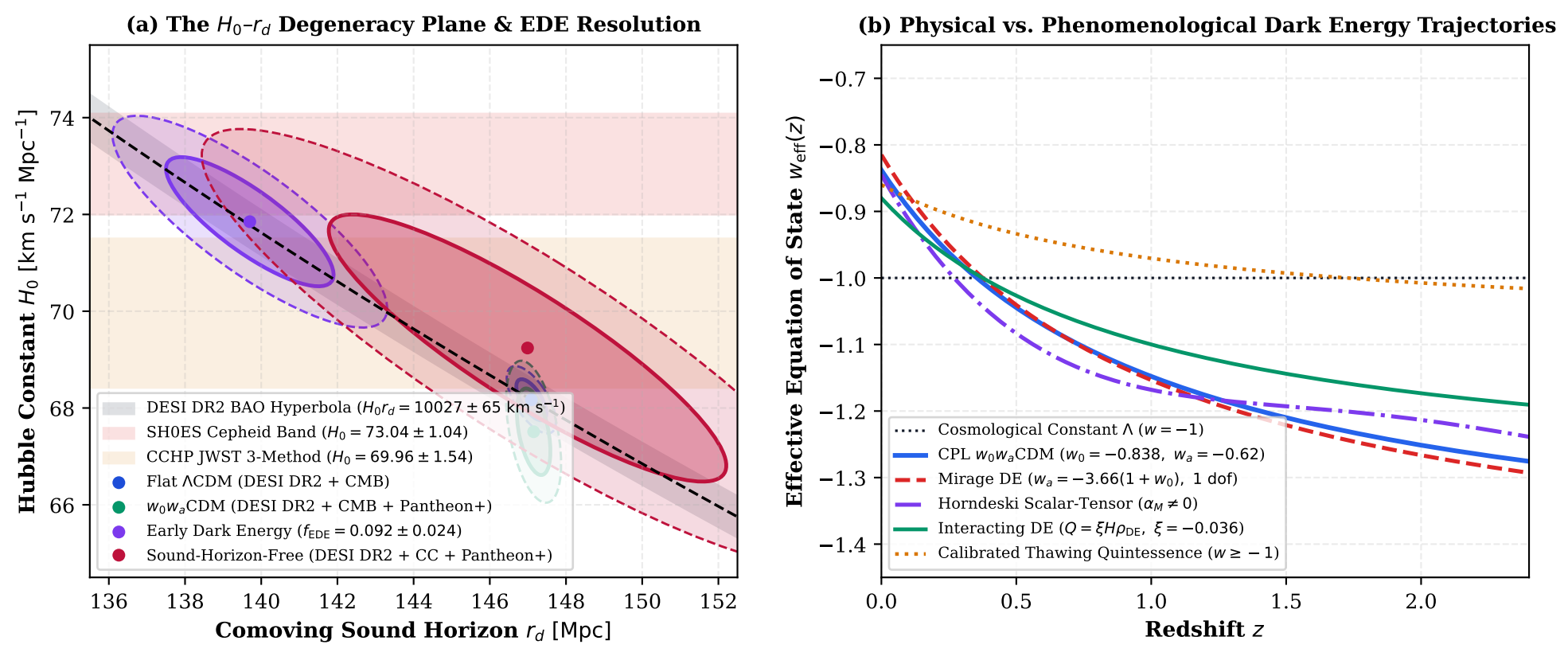

(a) The $H_0$–$r_d$ Degeneracy Plane & EDE Resolution
Hubble Constant $H_0$ [km s$^{-1}$ Mpc$^{-1}$]
Comoving Sound Horizon $r_d$ [Mpc]
74
72
70
68
66
136
138
140
142
144
146
148
150
152
DESI DR2 BAO Hyperbola ($H_0 r_d = 10027 \pm 65$ km s$^{-1}$)
SH0ES Cepheid Band ($H_0 = 73.04 \pm 1.04$)
CCHP JWST 3-Method ($H_0 = 69.96 \pm 1.54$)
Flat ΛCDM (DESI DR2 + CMB)
$w_0 w_a$CDM (DESI DR2 + CMB + Pantheon+)
Early Dark Energy ($f_{\rm EDE} = 0.092 \pm 0.024$)
Sound-Horizon-Free (DESI DR2 + CC + Pantheon+)
(b) Physical vs. Phenomenological Dark Energy Trajectories
Effective Equation of State $w_{\rm eff}(z)$
Redshift $z$
−0.7
−0.8
−0.9
−1.0
−1.1
−1.2
−1.3
−1.4
0.0
0.5
1.0
1.5
2.0
Cosmological Constant Λ ($w = -1$)
CPL $w_0 w_a$CDM ($w_0 = -0.838$, $w_a = -0.62$)
Mirage DE ($w_a = -3.66(1 + w_0)$, 1 dof)
Horndeski Scalar-Tensor ($\alpha_M \neq 0$)
Interacting DE ($Q = \xi H \rho_{\rm DE}$, $\xi = -0.036$)
Calibrated Thawing Quintessence ($w \geq -1$)